

SIH26103 — MoSPI PAIMANA Project Monitoring Platform

Final Consolidated Solution Document

1. Problem Statement — Core Ask

MoSPI's Infrastructure & Project Monitoring Division (IPMD) monitors Central Sector Infrastructure Projects (₹150 Cr+) via the PAIMANA portal (evolved from OCMS since 2006). Historical data exists (cost, expenditure, progress, milestones, delays) but is used mainly for reporting/charts, not for predictive risk analysis.

Goal: Move from reactive monitoring ("what already happened?") to proactive monitoring ("what is likely to happen next, and which project needs attention now?").

2. Real Problems Identified (from ground research + PS)

1. Project's actual progress not accurately/timely known
2. Delay is discovered only after it has already happened
3. Cost overruns
4. Time overruns
5. Comparing different projects is difficult (no benchmarking)
6. Root cause of a problem is hard to identify
7. Large historical data is underutilized
8. Officers must decide priority among thousands of projects
9. Billing/commission-driven corruption — inflated bills and quantities via Contractor → Petty Contractor → Engineer commission chains

3. Solution — Point-by-Point

3.1 Actual Progress Tracking

- Daily Record module — full project lifecycle log, date-filterable
- Project base profile: size, location, cost, duration, contractor
- Geo-tagged photo + material photo uploads by Site Engineer
- Material bills uploaded and verified by inspection team

- Second module: Documentation page (officer-submitted docs, MoSPI-only access)
- Role-based reminders (Site/Sub/Material/QA-QC Engineers) with visit-tracking
- Missed visits trigger automatic signal to MoSPI team

3.2 Delay Detection (Reactive → Proactive)

- Every "paused/off" status requires a mandatory reason from daily record
- System flow: Problem starts → Abnormal pattern detected → Early Warning → Officer investigates → Intervention
- Delay factors captured directly at source (labour, material, tools, land/legal issues)

3.3 Cost Overrun Control

- AI computes an expected cost/material estimate from daily record data (work done vs material used)
- Actual billing entries compared against this estimate — mismatches auto-flagged
- MoSPI team approval gate required for flagged bills with mandatory reason

3.4 Time Overrun Control

- Progress trend analyzed against deadline trajectory
- Automatic warning to officer/manager when current pace won't meet deadline

3.5 Benchmarking Across Projects

- Multi-site dashboard for managers
- Comparison page: progress/cost/duration/contractor across similar-profile projects
- Example insight: "Project A is 70% complete; similar projects average 82% at this stage"
- View-only (no edit) — pure statistics display

3.6 Root-Cause Identification

- Predefined reason taxonomy: land acquisition, funding, contractor performance, material availability, weather, litigation, design/scope change, etc.
- Reason auto-routed to the responsible role (e.g., material issue → Material Engineer)
- Non-response triggers escalation with full timestamped record

3.7 Historical Data Utilization (Core ML Engine)

- Historical project data (planned vs actual progress, expenditure, milestones) used as ML training data
- Model learns patterns correlated with past cost/time overruns
- Ongoing project's current data compared against learned patterns
- Output: Risk probability (cost overrun %, time overrun %) + composite Risk Score (0-100)

- Monthly risk-trend graph acts as a visual early-warning signal
- Key pattern example: expenditure progressing faster than physical work = high-risk indicator

3.8 Corruption / Billing Mismatch Risk

- Full contractor hierarchy mapped: Client → Main Contractor → Petty Contractor → Labour
- Corruption vector: commissions to Site/Material/Billing/QA-QC Engineers inflate bills & quantities
- Solution: cross-check daily ground data (material used, work done, photos) against submitted bills
- AI-estimated expected cost vs actual bill mismatch = anomaly flag → mandatory reason → MoSPI approval

3.9 Data Accessibility & Officer Priority

- LLM Assistant: officers/admins query in natural language ("why is Project X delayed?") and get a synthesized answer from reasons + risk data
- Priority Queue: auto-ranked worklist of projects needing urgent attention (by risk score, days-pending)

4. Gaps Closed (Round 2 Refinement)

Gap Resolution

Officer Priority Queue Dedicated ranked worklist — top N high-risk projects with reason + days-pending, sorted automatically

Actual dataset question Historical/ML part uses a synthetic PAIMANA-style dataset (planned vs actual progress, expenditure, milestones) for training. Daily Record part is framed as a new future data-collection layer (doesn't exist in MoSPI today) — kept clearly separate to avoid confusion

Photo-based AI verification Simplified — full computer-vision stage/quantity detection dropped as unrealistic for hackathon scope. AI role limited to flagging quantity/cost mismatches via calculation, not image-based quantity extraction

Actionable recommendations Every risk alert now includes a suggested next action (e.g., "Land clearance is top risk driver → escalate to concerned department"), generated from explainability output

PAIMANA/OCMS integration story One architecture slide: system sits as an AI layer on top of PAIMANA, consuming existing data via API, with Daily Record as an add-on module

Success/evaluation metric Demo includes a before/after comparison: traditional delay-detection lag vs system's early-warning lead time, plus a model accuracy/precision figure on historical test data

5. Final Consolidated Feature List

1. Daily Record Module (progress, photos, material, billing, role-based flow, reminders)
2. ML Risk Engine (cost/time overrun probability, composite risk score, risk trend graph)
3. Root-Cause + Explainability (top contributing factors, reason taxonomy)
4. Actionable Recommendations (per alert)

5. Officer Priority Queue (auto-ranked worklist)
6. Benchmarking Dashboard (cross-project comparison, view-only)
7. Billing Mismatch Detection (expected vs actual, approval gate)
8. LLM Assistant (natural-language querying)
9. PAIMANA/OCMS Integration Layer (conceptual architecture)
10. Impact/Evaluation Metric (early-warning lead time + prediction accuracy)

6. Final UI Structure

Roles

1. MoSPI Admin/Team — full access, approvals, oversight
2. Field Officers (Site Engineer, Sub Engineer, Billing Engineer, QA/QC, Material Engineer) — daily submissions
3. Project Manager/GM (optional) — assigned-project view

Pages & Flow

Login / Role Selection

- Role-based redirect to appropriate dashboard

MoSPI Admin — Main Dashboard

- Summary cards: Total Projects, High-Risk Count, Delayed Projects, Budget vs Spent
- Priority Worklist widget (top 5–10 urgent projects)
- Portfolio-wide risk trend mini-graph
- Filters: sector, state, status
- Sidebar: Dashboard | Projects | Priority Queue | Benchmarking | Billing Alerts | Documents | LLM Assistant | Settings

Projects List Page

- Table/card view: name, sector, location, contractor, % progress, risk score (color-coded), status
- Search + filters → opens Project Detail Page

Project Detail Page (tabbed)

- Overview: basic info, planned vs actual progress bar
- Daily Record: date-filterable log, photos, material, work status, visit log
- Risk & Prediction: risk gauge, trend graph, top factors (explainability chart), recommendation box
- Billing: submitted vs estimated bills, flagged mismatches, approve/reject/ask-reason actions

- Documents: officer-uploaded files
- Comparison/Benchmark: overlay vs similar projects

Priority Queue Page

- Ranked list: rank, project, risk score, primary reason, days-since-flagged, action button

Benchmarking Page

- Filter by sector/size/region, side-by-side comparison charts, view-only

Billing Alerts Page

- All flagged mismatch cases across projects, status (Pending/Approved/Rejected), reason history

LLM Assistant Page (or floating chat widget)

- Natural-language Q&A tied to project data

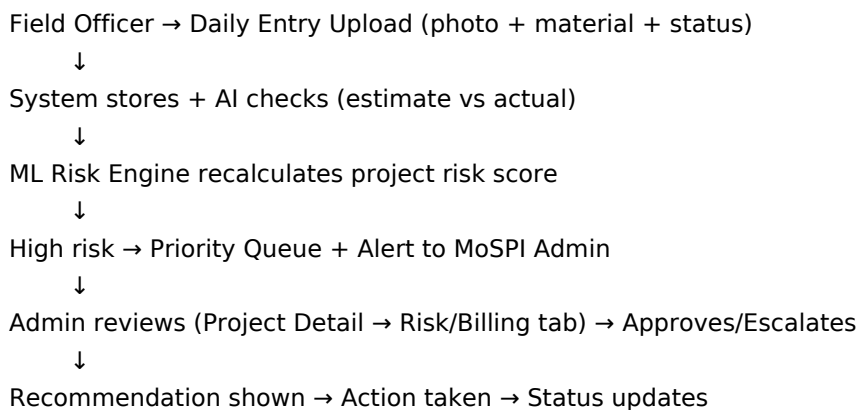
Field Officer Side (simplified view)

- My Assigned Projects
- Today's Task (reminders)
- Daily Entry Upload form (photo + geo-tag, material quantity, status, notes)
- My Documents

Reports/Export Page (optional, if time permits)

- Generate portfolio/project summary report

End-to-End Flow



7. Status

Finalized for the college/internal round (idea + presentation stage). Full scope intentionally kept broad to demonstrate depth of thinking; can be trimmed to an MVP (3–4 core functional screens) if a working prototype demo becomes mandatory later.